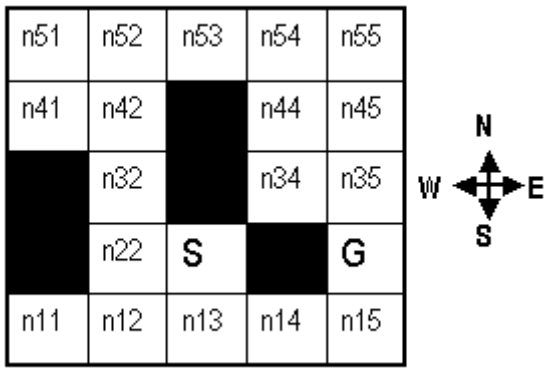
Maze

Consider the maze shown in the figure below. **S** is the initial state and **G** is the goal state. Dark squares are blocked cells. All squares, except for the start and goal states, are labeled, with row and column numbers. For example, n54 is the cell in the fifth row, fourth column.



- the graph states are the squares of the maze.
- there are links along the cardinal compass directions to neighbouring squares (that are not blocked). That is, diagonal moves are not permitted, nor are moves into or across a black square.
- the successors of a state are to be generated in a FIXED order to be drawn from left to right, namely N, E, S, W.
- We say that a state is **expanded** when a path to that state is pulled off of Q.

Find a path from **S** to **G** by applying the **A* algorithm with the Strict Expanded List, and the Manhattan distance heuristic**, indicate the order in which states are expanded. Enter a sequence of state names starting with S and ending with G, e.g. `S n13 n22 ... G`.